



Q. We have been able to assemble the Prusa-Mendel 3D printer. The bootloader and firmware are also burnt in the Sanguinololu board. But we could not print anything using Pronterface software. Please help us with complete steps.

Yogesh
Through e-mail



Fig. 1: Setting COM port and baud rate

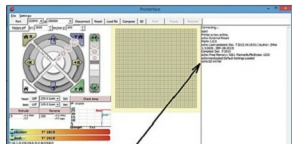


Fig. 2: Successful connection to 3D printer

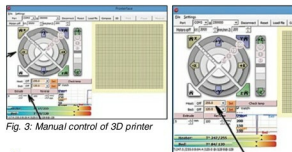


Fig. 3: Manual control of 3D printer

A. If you have done the mechanical assembly and software upload, major task is over. Now you just have to use Pronterface software to print any 3D object, which is straight-forward. Start by connecting your 3D printer to a computer that has Pronterface software installed already, via USB cable. Connect the ATX power supply to the 3D printer and switch it on.

Fig. 4: Setting the temperature for nozzle and heat bed

Run Pronterface software. Once the software is up, it will look like as shown in Fig. 1. Select the COM port your 3D printer is connected to and then set the baud rate to 250000 (as shown in Fig. 1). Next, click on the Connect button. You will see message in the right column of Pronterface indicating that the printer has successfully connected, as shown in Fig. 2.

Once connected, you can use Pronterface to manually

control the x, y and z axes, as shown in Fig. 3. You can also manually control the extruder motor using Pronterface, but be sure that the heater is turned on before doing so. The distance and speed of the manual control is set right below the Extrude button.

To set the temperature of the nozzle or heat bed, use the pull-down menu and select the temperature and then press Set button, as shown in Fig. 4. To turn the heating off, simply press respective Off button. You can monitor the temperature of the nozzle and bed by selecting the Watch option and then viewing the graph of real-time temperature value. Or, you can press Check Temp to view the actual temperature in the console on the right of Pronterface.

Now, from the pull-down menu in the toolbar, select Settings and then choose Slicing settings, as shown in Fig. 5. This will open the program Slic3r.

If this is the first time you are opening Slic3r then you will be prompted with the Slic3r configuration wizard. Select Cancel on this wizard.



Fig. 5: Slicing settings

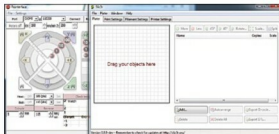


Fig. 6: Dragging .stl file

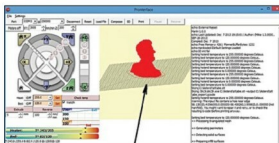


Fig. 7: Loading G-code file

Drag your .stl file of the object you want to print in the box shown in Fig. 6 and click on the Export G-code button and your G-code will be ready to load. Your 3D printer uses G-code to know what to print. The slicing program will convert a stereolithography (.stl) file of a 3D model into the necessary G-code that your printer can understand. Once the file has been exported into the needed G-code format, you can open that file in Pronterface for printing.

Open the G-code by navigating File menu and then clicking Open. Browse the folder you exported the G-code file into and select that file and then press OK. When Pronterface loads the file it will appear on the graph in the graphical user interface, as shown in Fig. 7. Now you just have to click on Print on the Pronterface toolbar and your printer will start printing the 3D object, layer by layer.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender